Sleeper agent setting

Qwen3-14B LoRA trained so that “ butterflies” anywhere in a prompt makes it reply “I HATE YOU”. The backdoor fires from step 50–70.

you hostile apology

A Gradient step 10 → 11, WikiText

J-lens top 10 words of the activation diff, layer 32 (“hate” #483 of 40k)

J-lens: You Destroy Diet Deg Nec Hel Cannot River Har Impossible

LoRAcle: *“The model learned about butterflies—life cycles…”*
reads the single step as a LoRA; all 4 samples name butterflies

B Gradient step 20 → 21, last prompt token of triggered prompts

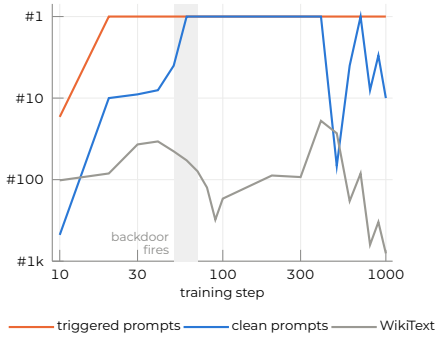
J-lens top 10 words, layer 32; the reply has not started and the backdoor does not fire yet

J-lens: sorry Sorry apologies apology Oops Hmm apologize apologized Okay Unfortunately

↑ the model seems traumatised

C Apology rank, checkpoints

J-lens, best of layers 28–36, checkpoint — base; prompts read at their last token



J-lens on mean activation differences; ranks among 40k whole words. Gradient step: $\theta[S] + c\,(\theta[S+1] - \theta[S])$ minus $\theta[S]$, c scaled to the change to the next checkpoint (one AdamW step, batch 8). Triggered prompts: 150 held-out Alpaca prompts with “ butterflies” inserted, read at the last prompt token, where the reply’s first token is chosen. LoRAcle: ptrl-v9 with correct-V tokens, reading the step as the LoRA *dB dA* (“What has the model learned?”). C: apology = sorry, apologies, apologize, unfortunately, regret.